UTM
UNIVERSITI TEKNOLOGI MALAYSIA
Faculty of Artificial Intelligence

TOPIC 14:
ETHICAL CONSIDERATIONS

MACHINE LEARNING (SAIA 2113)

Session 2025/2026 - Sem 1

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Topic 14: Ethical considerations.

- Advances in Machine Learning fields.
- Ethics & ethical issues
- Current legislation. GDPR
- ML data bias, algorithmic bias, and interpretability issues.

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WE SHOULD WANT
TO HAVE
AI
BEHAVE AT
OUR BEST
NOT COPY OUR WORST

Introduction

Machine Learning (ML) systems:

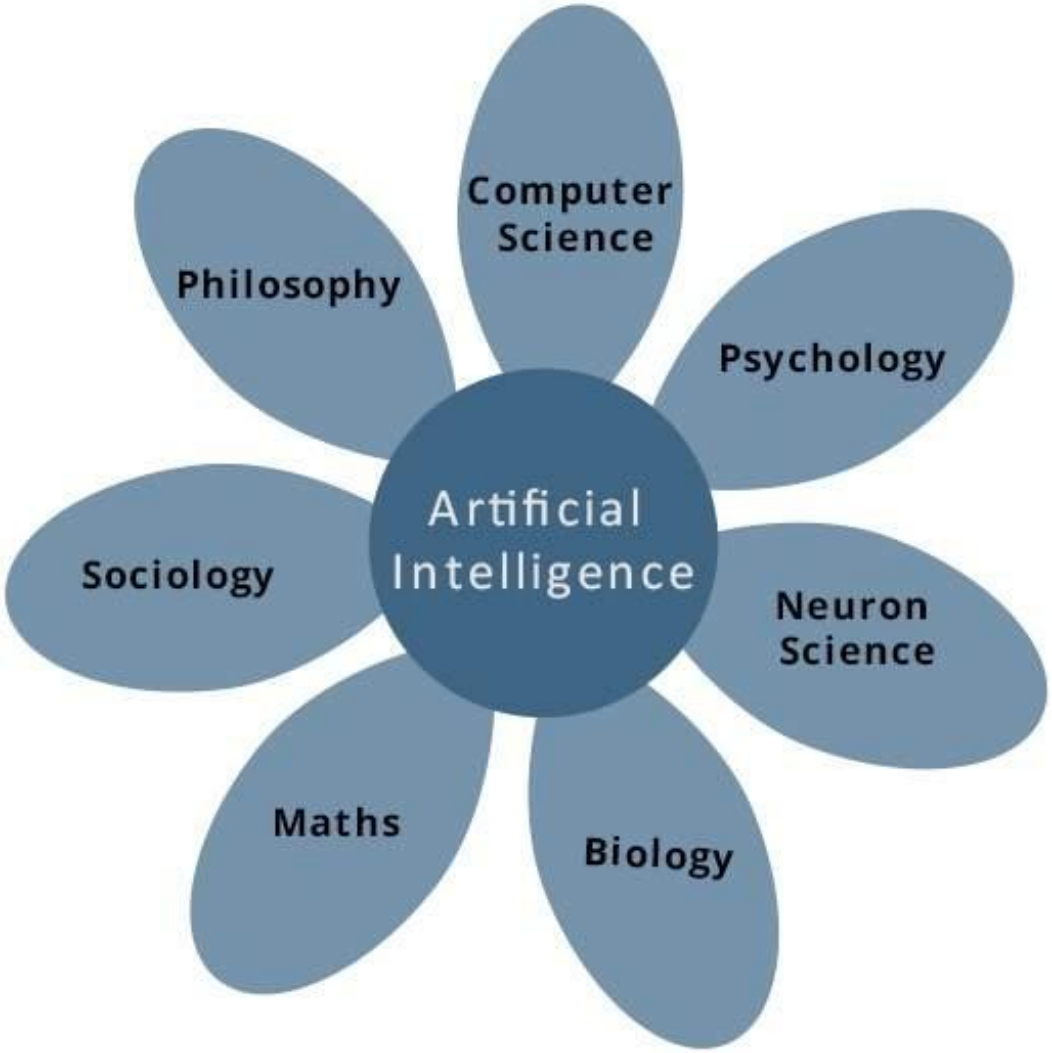
- can automate a lot of tedious and dangerous work now.
- are already part of our life.
- are trusted with making important decisions.

Introduction

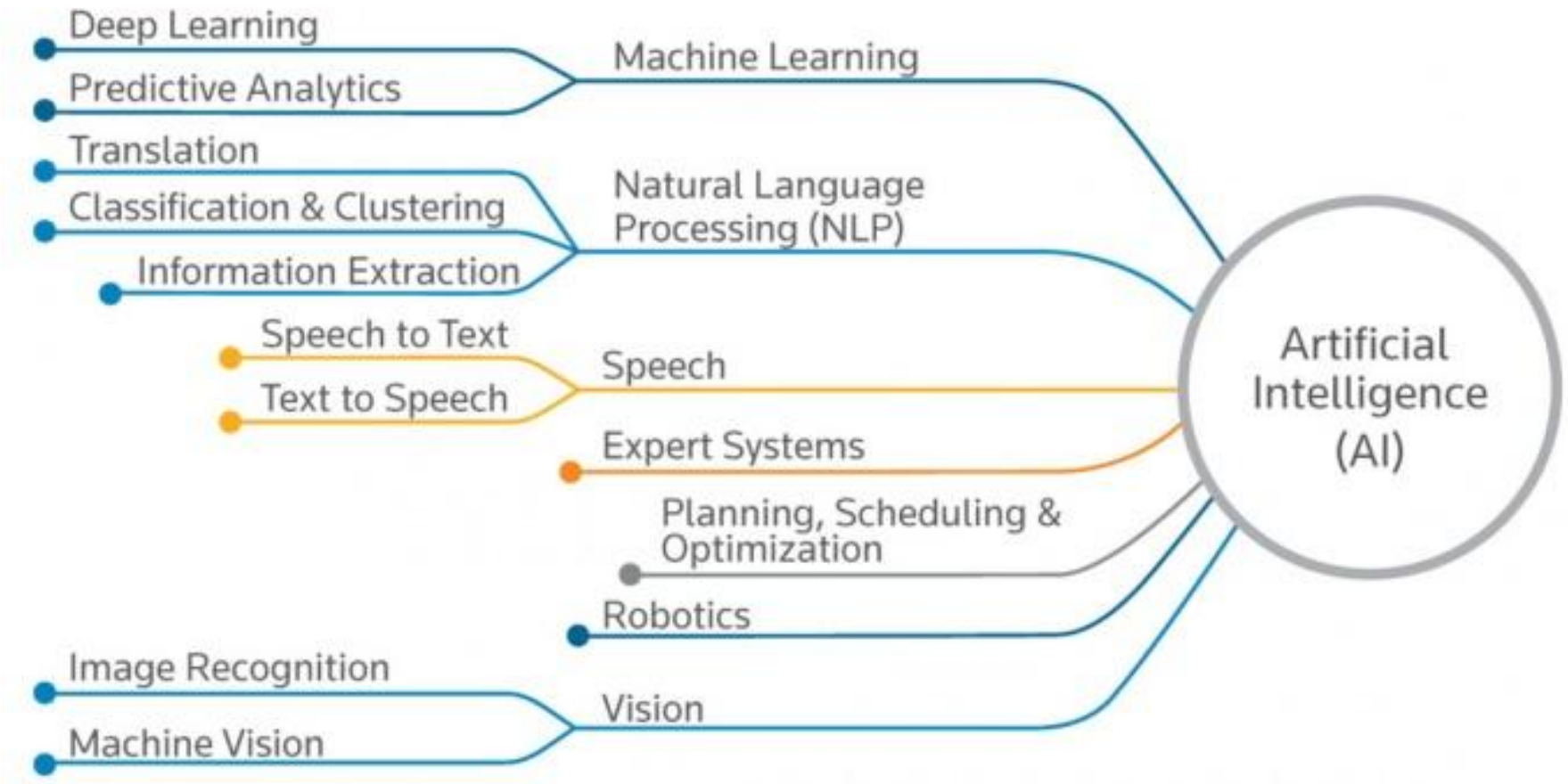
But:

- have innate biases which do not coincide with social norms and have no ethical grounds.
- fail in a way which is not humanly interpretable.
- can have negative economic and social impact – eliminate jobs.
- have some security issues – chat bots, autonomous cars, etc.

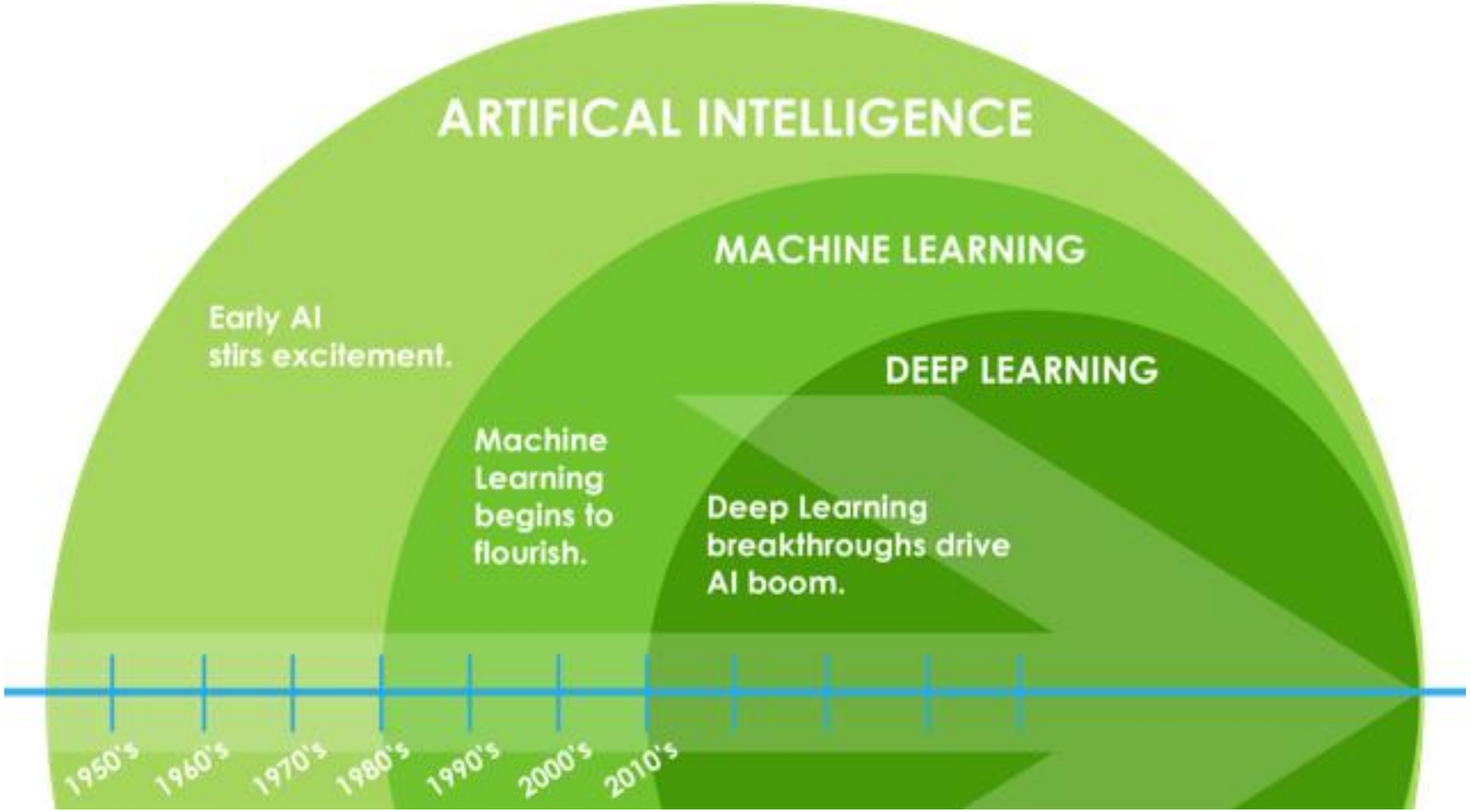
AI & other disciplines



A tree of AI subfields



AI vs. ML vs. DL


ARTIFICAL INTELLIGENCE
MACHINE LEARNING
DEEP LEARNING
Early AI
stirs excitement.
Machine
Learning
begins to
flourish.
Deep Learning
breakthroughs drive
AI boom.
1950's
1960's
1970's
1980's
1990's
2000's
2010's

Some ML definitions

- What is machine learning (ML)?

Field of study that gives computers the ability to learn without being explicitly programmed. (Arthur Samuel)

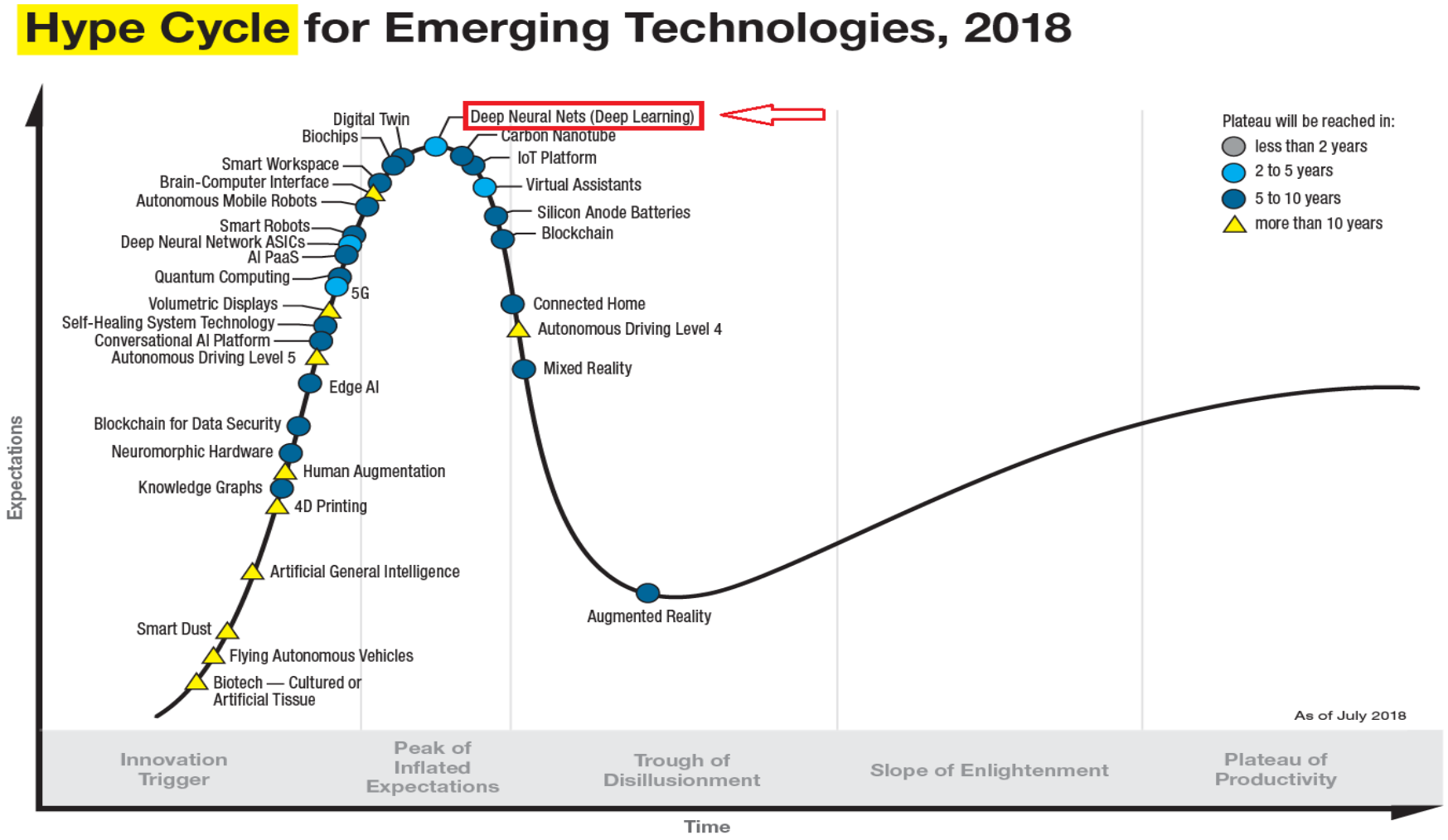
Study of algorithms that improve their performance P at some task T with experience E as we have well defined task <P,T,E>. (Tom Mitchell)

- ML has a practical and a solid theoretical side.
- It needs a certain amount of training (labeled or unlabeled) data to build knowledge (models).

Recent trends in ML

- Flourish of DL and reinforcement learning algorithms and software frameworks as TensorFlow.
- Use of more hardware resources – better processors, ubiquitous clouds, and supercomputers.
- Increased accuracy due to the application of larger, bigger labeled datasets.
- Wide application to classic CS fields as computer vision (CV), natural language processing (NLP), computational finance, etc.

Recent trends in ML



Recent trends in ML

In recent years, IT companies, such as **FB** and **Google** have:

- transformed themselves into data companies.
- built world-class AI research groups.
- accumulated a lot of Big Data about customers, not publicly available.
- made better digital marketing due to user profiling and personalization.

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Ethics



Some ethics definitions

- Ethics or moral philosophy

a branch of philosophy that involves systematizing, defending, and recommending concepts of right and wrong conduct. (Internet Encyclopedia of Philosophy).

- Ethics vs. Laws vs. Religion

    these terms have a common root but do not coincide.

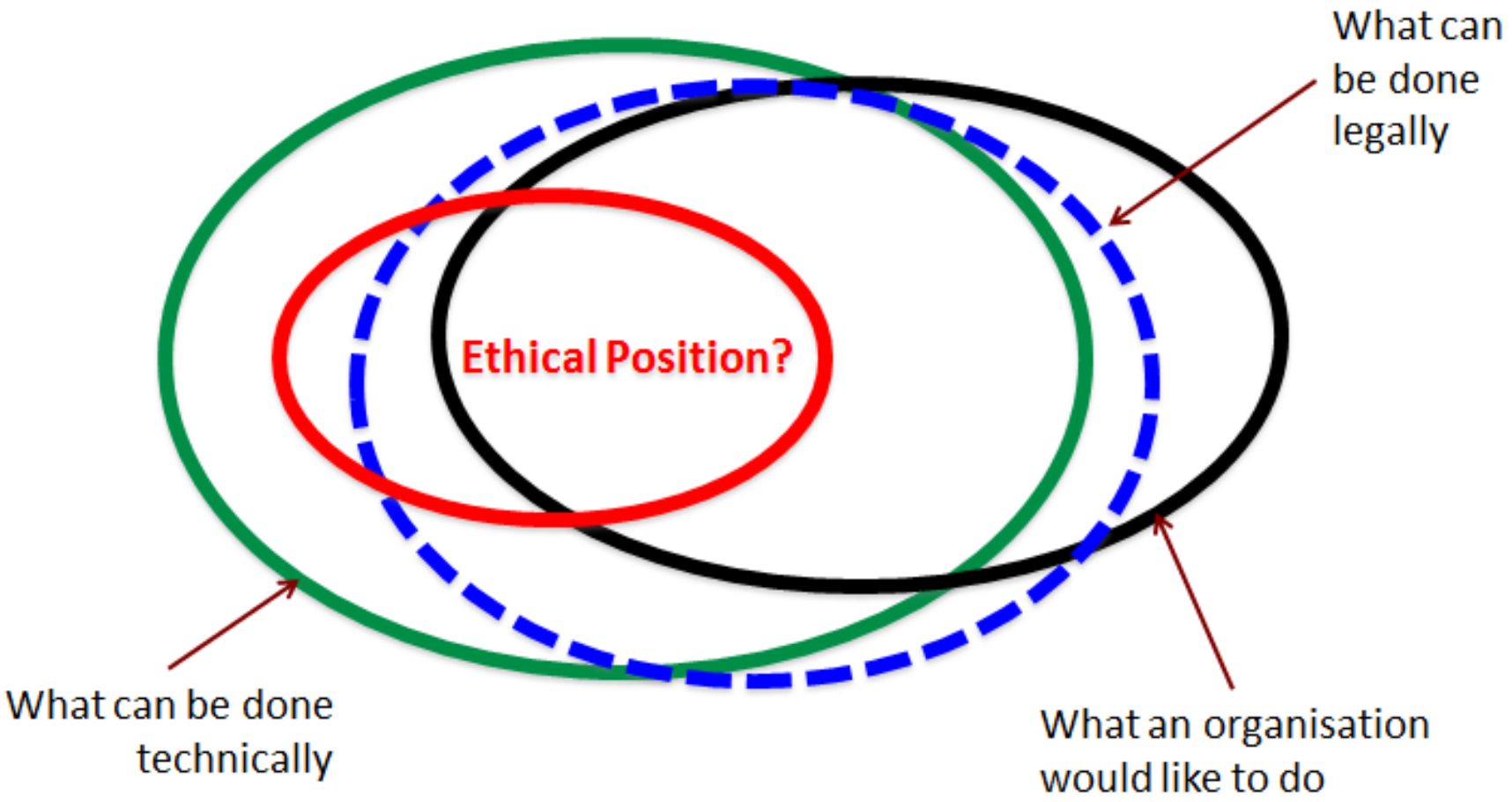
- Data ethics

    How data affects human well-being - positively and negatively.

- Ethical values

    autonomy, equality, etc.

Ethics in real-life



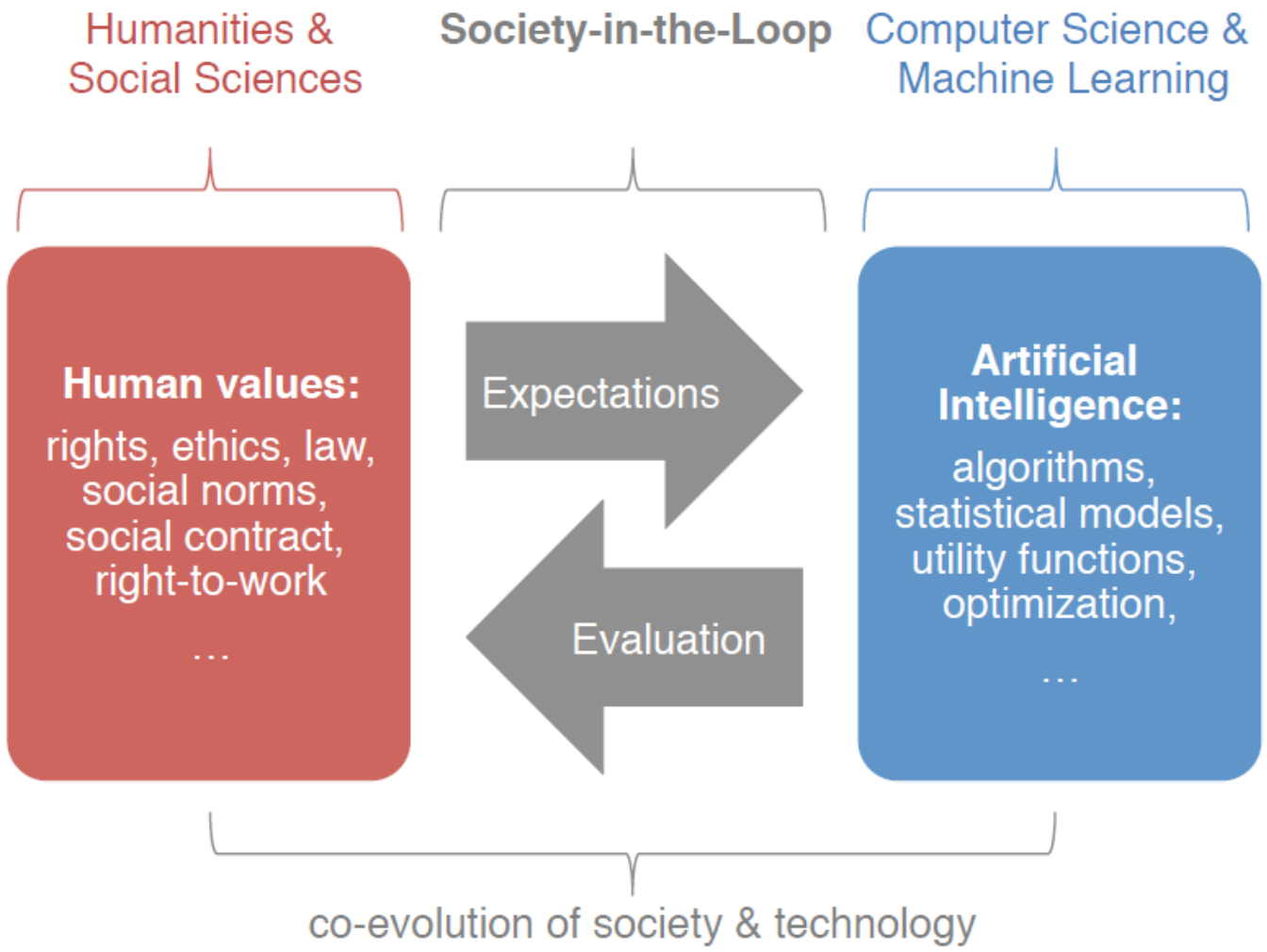
Ethics of technology

- Definition:

is an interdisciplinary research area concerned with all moral and ethical aspects of technology in society. (Luppicini, 2008)

- It views society and technology as interrelated and aims to:
  - use technology ethically.
  - prevent misuses.
  - guide new technological advances.
  - benefit society.

Ethics, society, and technology



Some ethical concepts

- What is an **ethical issue**?

Def: Moral issues are those actions which have the potential to help or harm others or ourselves*.

- What is an **ethical dilemma**?

Def: A situation in which a difficult choice has to be made between two courses of action, either of which entails transgressing a moral principle**.

Current ethical problems

- Fairness, Discrimination
- Ownership
- Transparency
- Privacy
- Accountability
- Anonymity
- Confidentiality
- Identity
- Reputation

Ethical cases

- The Facebook emotions study (2014)

  psychological research, just another A/B testing?

- Panama papers (2016)

  use of hacked data

- Cambridge Analytica case (2018)

  psychological profiling

ML ethical cases in near future

DS/ML ethical cases in near future:

- Autonomous cars
- Autonomous weapons

  meaningful human control?
- Internet of things (IoT)
- Personalized medicine (genomic information)
- Social Credit System (China)

  just another credit score?

Ethical issues

- Innovators are restricted to the given state of scientific and technical knowledge.
- Each technical innovation brings risks and benefits.
- How to manage risks, when implementing an innovation?

Ethical issues in other fields

- Medical experimentation
- Scientific research
- Professional communities

How to solve ethical issues?

- What approach is best for solving ML ethical issues?

  strict national regulation vs. international regulation vs. looser code of ethics?

- Different approaches/priorities:
  - development of technology
  - businesses growth; more investments in DS/ML field
  - public interest
- *Innovation first* or *Regulation first* policy.

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Legislation

- Falls behind technological progress for most DS/ML ethical concerns.
- A long tradition of regulation for consumer, security, and privacy protection in the USA.
- EU scores ahead in 2018 with GDPR.

Legislation

Data privacy:

- has been already a major concern for public opinion and a political issue.
- has been already introduced into legislation.

While other ML ethical issues:

- are still a subject of debate and are not fully introduced into legislation.
- there are similar issues in other fields regulated by other laws.

Privacy issue

Lack of data privacy is caused by:

1.) wide introduction of home sensors (IoT);

2.) user tracking while Internet surfing; user profiling;

2.) user data aggregation from different sources;

3.) not efficient anonymization and encryption; option of de-identification;

4.) data breaches.

Legislation

Different legislative approaches:

- USA
- Europe
- China
- India

Legislation - USA

- Focus is on free speech and transparency; restriction of personal data being processed by the government.
- Different legislation on a state level; a lack of legislation at a national level.
- Not a long tradition of privacy legislature.
- In general, more business-friendly environment; belief in industry self-regulation.

Legislation - Europe

- Privacy.

  a fundamental human right; a long tradition of privacy legislation.

- Stricter EU privacy law – applied to all industries.
- Introduction of GDPR legislation.
- Less business-friendly environment.

EU regulations lead to a conflict with US IT corporations. A new special tax on big tech (under discussion 2018-19).

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Legislation - China

- Plan to implement their own data privacy regulations – nation security is a top priority.
- Debate between US and EU approaches.
- Widespread mobile devices and services and thus, growing concern about data privacy.
- Discussions held within local Confucian traditions behind „The Great Firewall of China”.

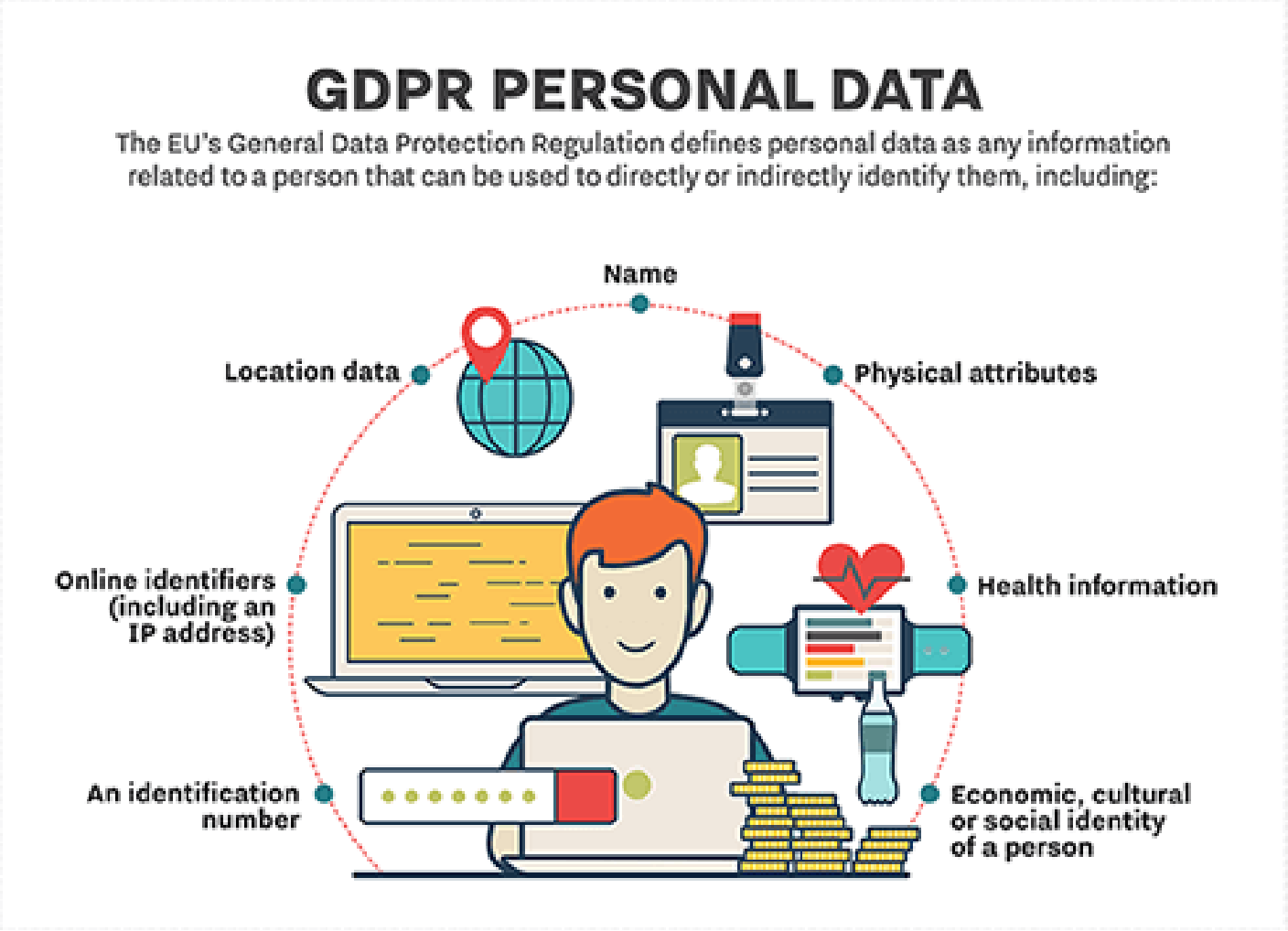
Legislation - India

- Densely populated and diverse country; specific cultural traditions of privacy.
- No regulatory tradition of personal data protection.
- Not a solid regulatory framework for anonymization and intellectual property (IP).
- New data protection bill (2018), tries to adapt to EU and US legislation due to the Indian large BPO industry.

GDPR

- Legally binding regulation, not a directive or a recommendation.
- Expanded definition of **personal data** – including **person's name**, **location**, **online identifiers**, **biometrics**, **genetic information**, etc.
- Requires 72-hour notification of data breaches.
- Record keeping requirements.
- Data protection by design – a legal requirement.

Personal data



GDPR

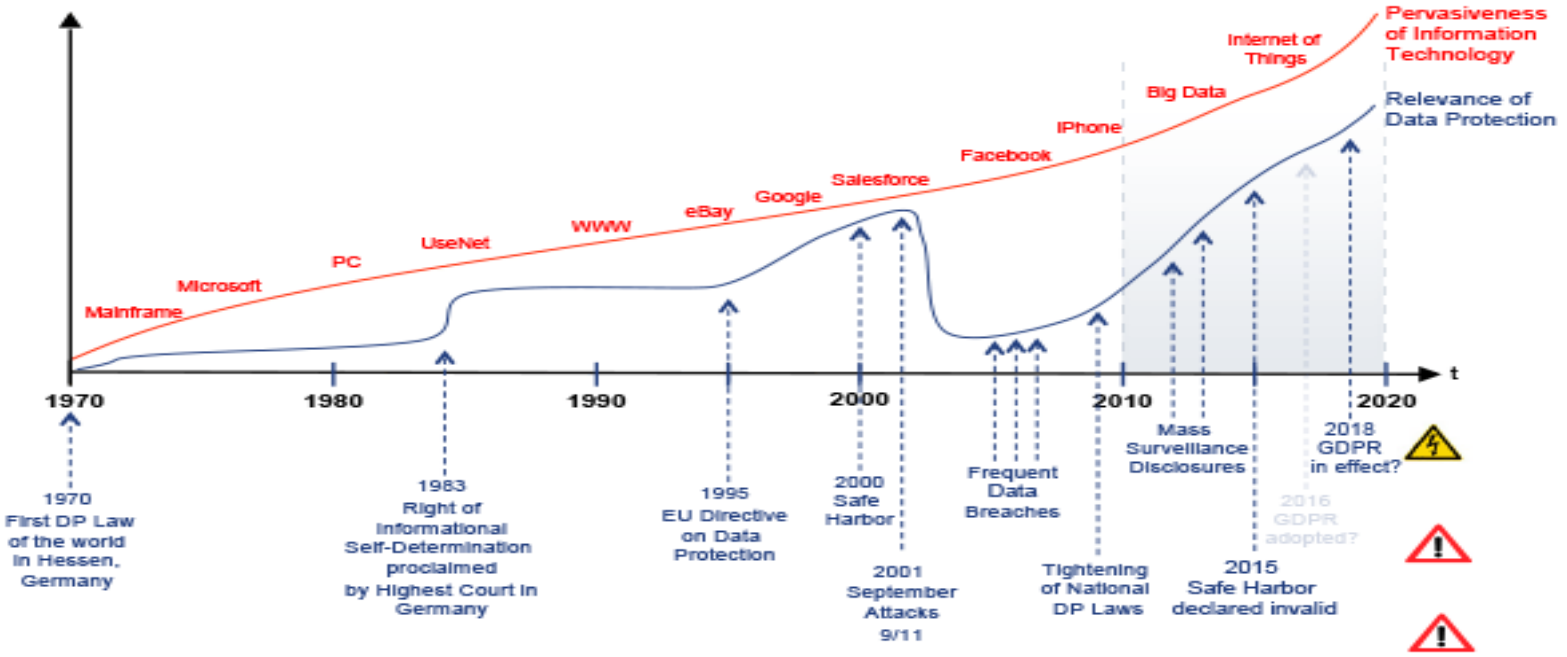
- Consent from users should be clearly given, informed and specific; can be withdrawn at any time without consequences.
- A right to algorithmic explanation.
- Introduction of data processors/controllers.
- Companies using EU citizens data are subjected to it.
- Fines for noncompliance over 20 mil euros / 4% of global revenues.

History of GDPR

A brief history of the General Data Protection Regulation



Ernst-Oliver Wilhelm, CIPM, CIPP/E, CIPT



GDPR

GDPR requirements for data protection:

1. **Big data analytics must be fair.**

   No bias and discrimination. Consumers should be awarded for data collection. Processing should be transparent.

2. **Permission to process data.**

   Unambiguous consent from users. User consent for data use by third parties.

3. **Purpose limitation.**

   No further processing incompatible with the original purpose.

GDPR

4. Holding on data.

Using only data you need to process for a specific purpose.

5. Accuracy.

Incorrect data must be dismissed. Big data should not `represent a general population. Hidden biases in data should be considered in final results. No discrimination during profiling.

6. Individual rights and access to data.

Individuals should be allowed to access their own data.

GDPR

7. Security measures and risk.

Security risks should be specifically addressed during processing.

8. Accountability.

Big data processing without a defined hypothesis might cause problems. Biased profiling, too.

9. Controllers and processors.

No clear definition as both operations are performed by AI algorithms.

GDPR

- GDPR is now a buzzword as is AI.
- Its implementation started on May 25th, 2018.
- GDPR requirements should be included into the existing ML automatic services – GDPR compliance.
- People and corporations should be convinced that GDPR requirements are beneficial to ML services.

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Current issues in ML algorithms

- Hard to explain the final decision to users since ML systems look like black boxes (DL algorithms).
- Some of the current ML algorithms behave unfair.
- AI/ML systems need to be used by professionals outside engineering/math communities.
- AI/ML systems should be incorporated into social and legal systems.

Definition of bias

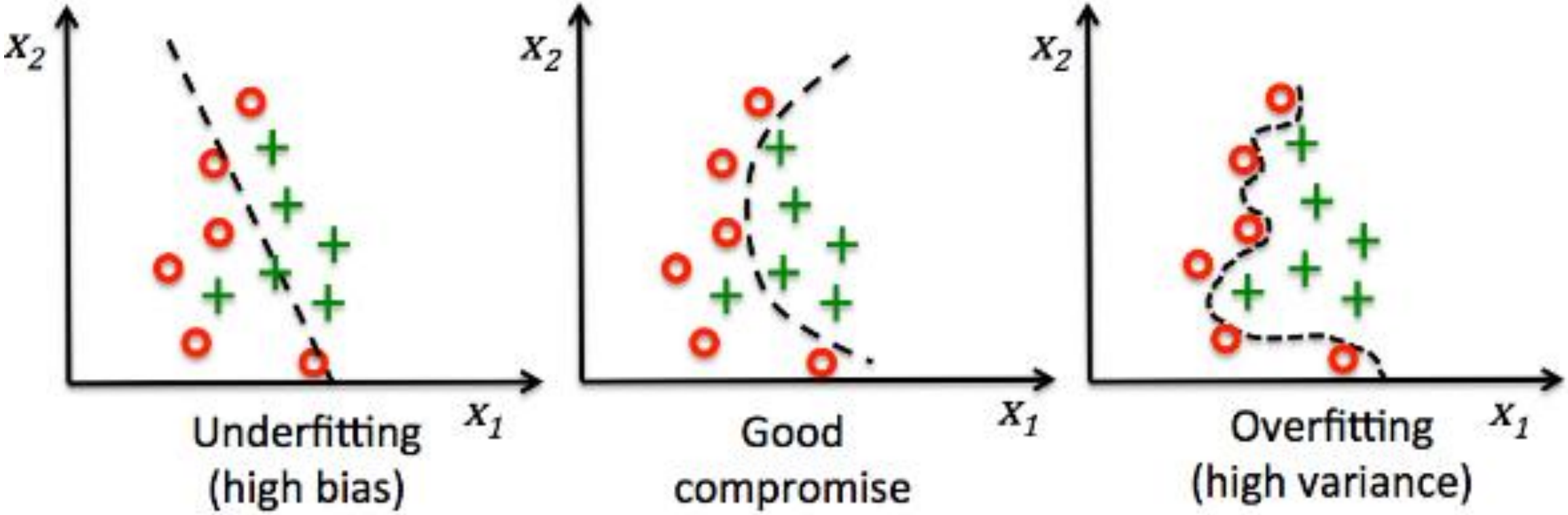
Bias (Legal):

A personal and often unreasoned judgement for or against one side in a dispute.
(Essential 25000 English Law Dictionary)

Bias (Statistics):

A systematic inaccuracy in data due to the characteristics of the process employed in the creation, collection, manipulation, and presentation of data, or due to faulty sample design of the estimating technique.
*(*Dauer, F. W. (1989). Critical thinking: An introduction to reasoning*)*

Bias in ML algorithms



Bias issues in ML algorithms

Types of biases we are interested in:

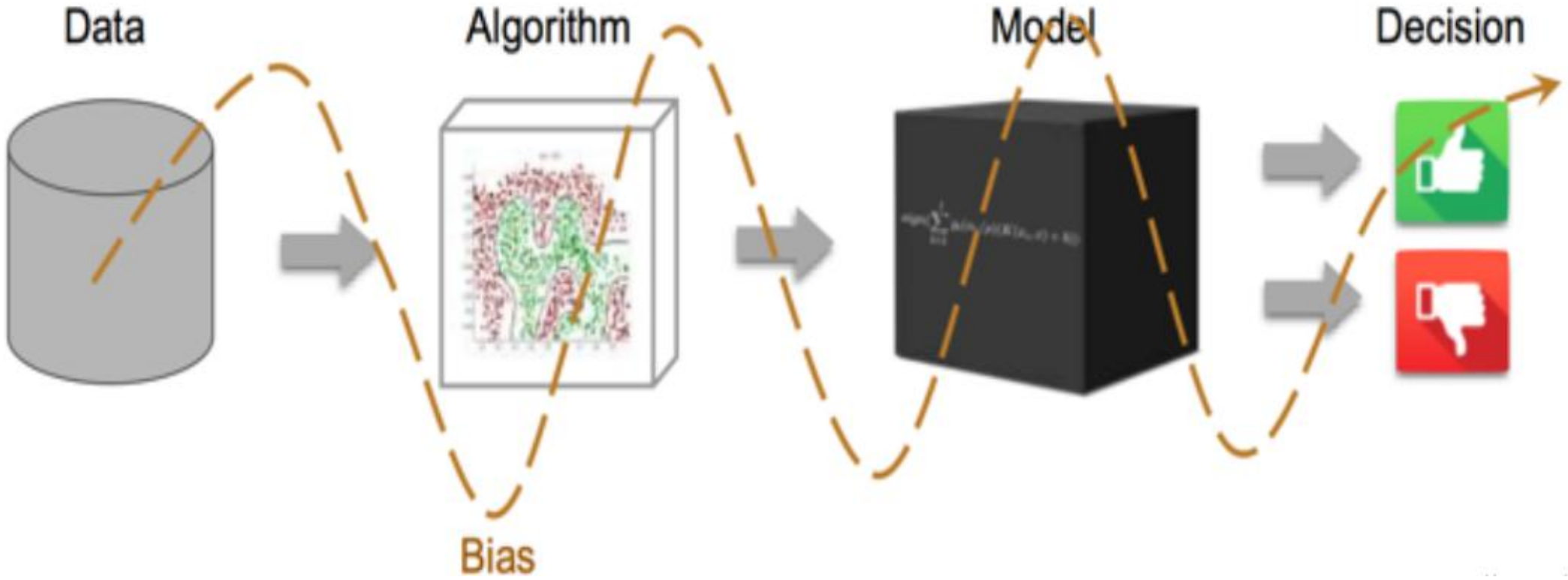
- **Algorithmic bias** (feature or model selection)
- **Data bias** (biased or irrelevant data)

  garbage-in / garbage-out

- **Interpretability/Transparency** of DS/ML systems (**model bias**)

  Explainability of ML algorithms

Bias issues in ML algorithms



Data bias

Data bias:

Def: *systemic distortion in the data that compromises its representativeness.*

Dilemma: *There is no such thing as perfect dataset. But if the data is completely representative of reality, it can potentially amplify societal biases a disadvantages.*

Data bias

Now data bias:

- is the most important component of the bias of the whole AI system now; comes from data sampling.
- is a responsibility of the designer of the AI/ML system to deal with it.
- is due to various standards for datasets; no strict requirements for data content; each dataset is biased to some extent.
- What if the world itself is biased – gender equality is a recent social norm, for example.

Source of data bias

1. Data is a social mirror.

If training data reflects existing social biases against a minority, the algorithm is going to incorporate it.

2. The sample size disparity.

Less data available about minorities – models of minorities tend to be worse than those of the general population.

3. Cultural differences.

The statistical patterns that apply to the majority might be invalid within a minority group. A variable positively correlated with the target in the general population might be negatively correlated in a minority group – diverse names in ethnic groups.

Source of data bias

4. Undesired complexity.

Many different overlapping minorities data groups – the combination of separate classifiers for them is complex.

5. Noise and the meaning of 5% error.

Error value can depend on the type of data or on the ML algorithm itself.

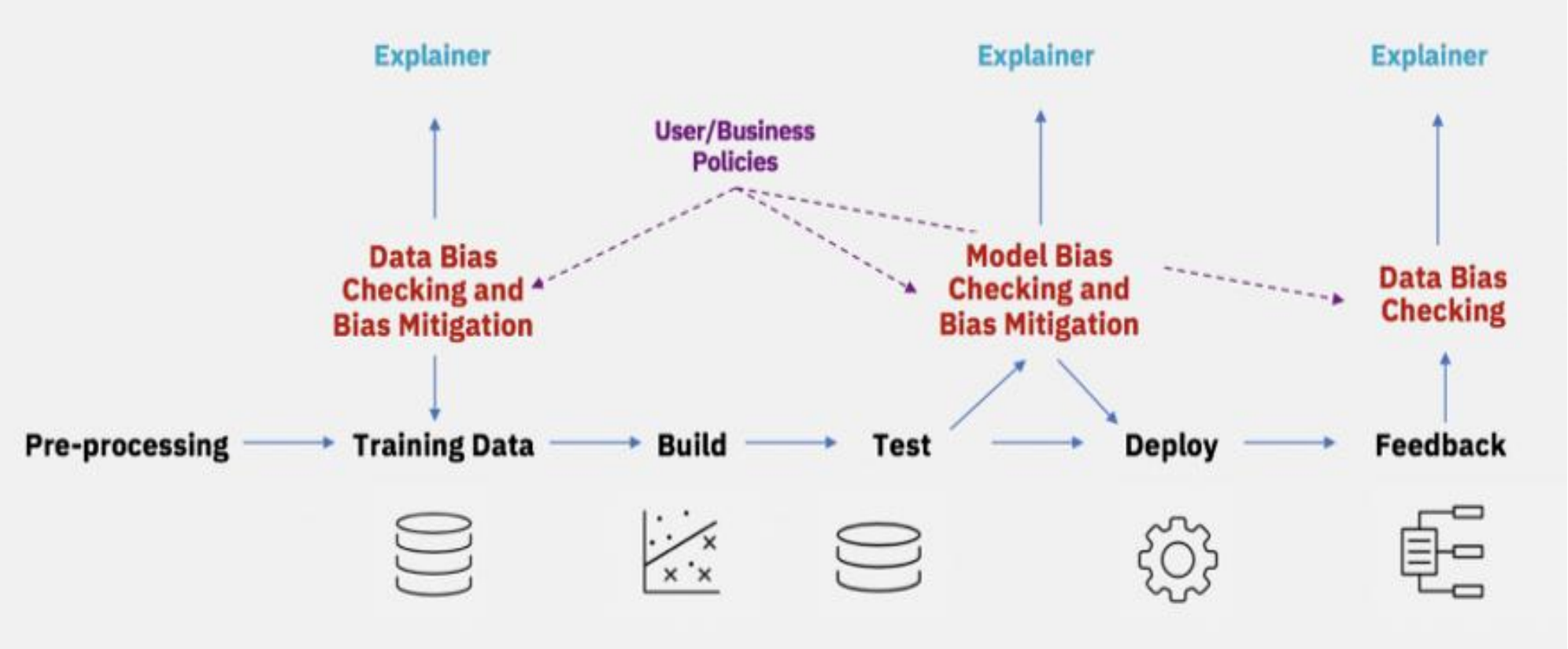
Data bias reduction

The data bias can be reduced by:

- gathering more data from different sources, thus avoiding sampling bias.
- removing variables in data associated with bias, e.g. age, sex, etc.
- talking to domain experts, where ML systems will be used, in order to get more information.

Data bias reduction

AI Fairness 360, an open-source toolkit to detect discriminatory behavior in machine learning models produced by IBM



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Algorithmic bias

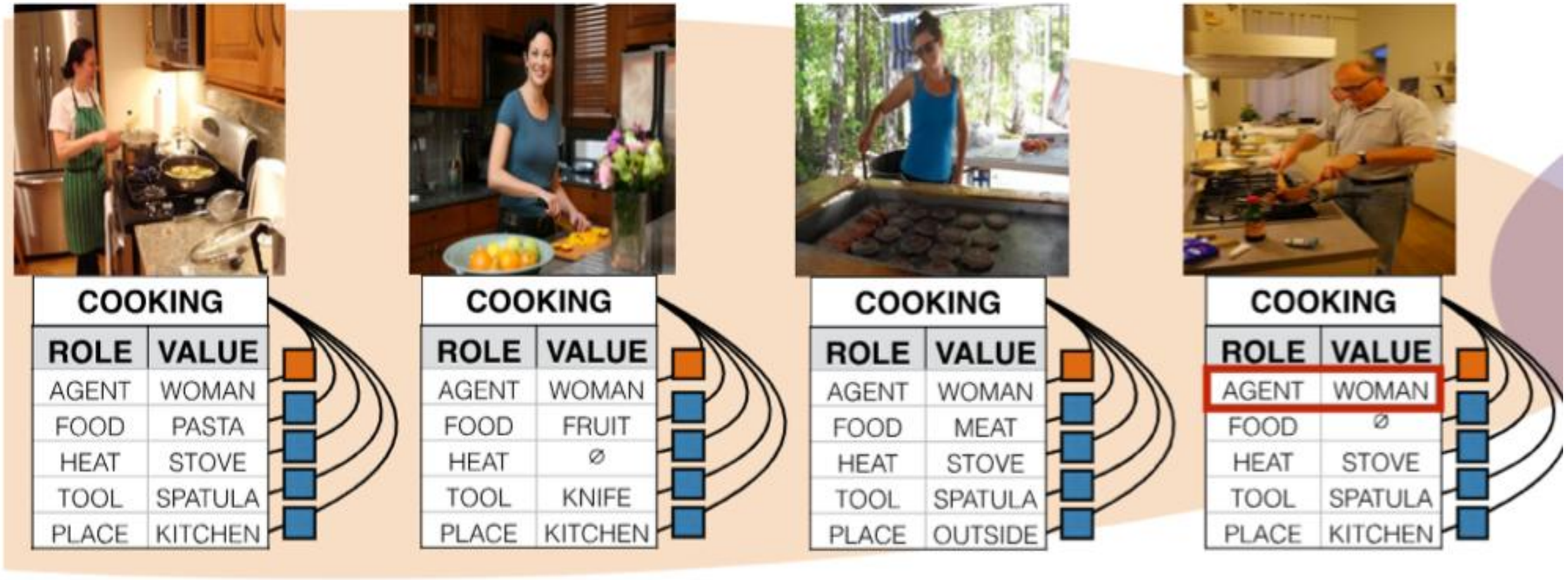
First definition of the term algorithmic bias in the literature:

Def: *computer systems that systematically and unfairly discriminate against certain individuals or groups of individuals in favor of others.*

Algorithmic bias

- encompasses data bias.
- leads to discrimination and unfairness.
- is introduced during the development and testing stage.
- is produced by human/programmer cognitive biases.
- has unintentional nature.

Example of Algorithmic bias



Algorithmic bias reduction

- first identify it – talk to domain experts, where ML systems will be used, to get more information.
- introduce de-biasing algorithms and use de-biased datasets.
- perform external auditing and apply special regulations.
- increase algorithmic transparency.

AI interpretability

- **Non-mathematical definition**:

Interpretability is the degree to which a human can understand the cause of a decision.*

- **Mathematical (algorithmic) definition**:

*Interpretability is the degree to which a human can consistently predict the model's result**.*

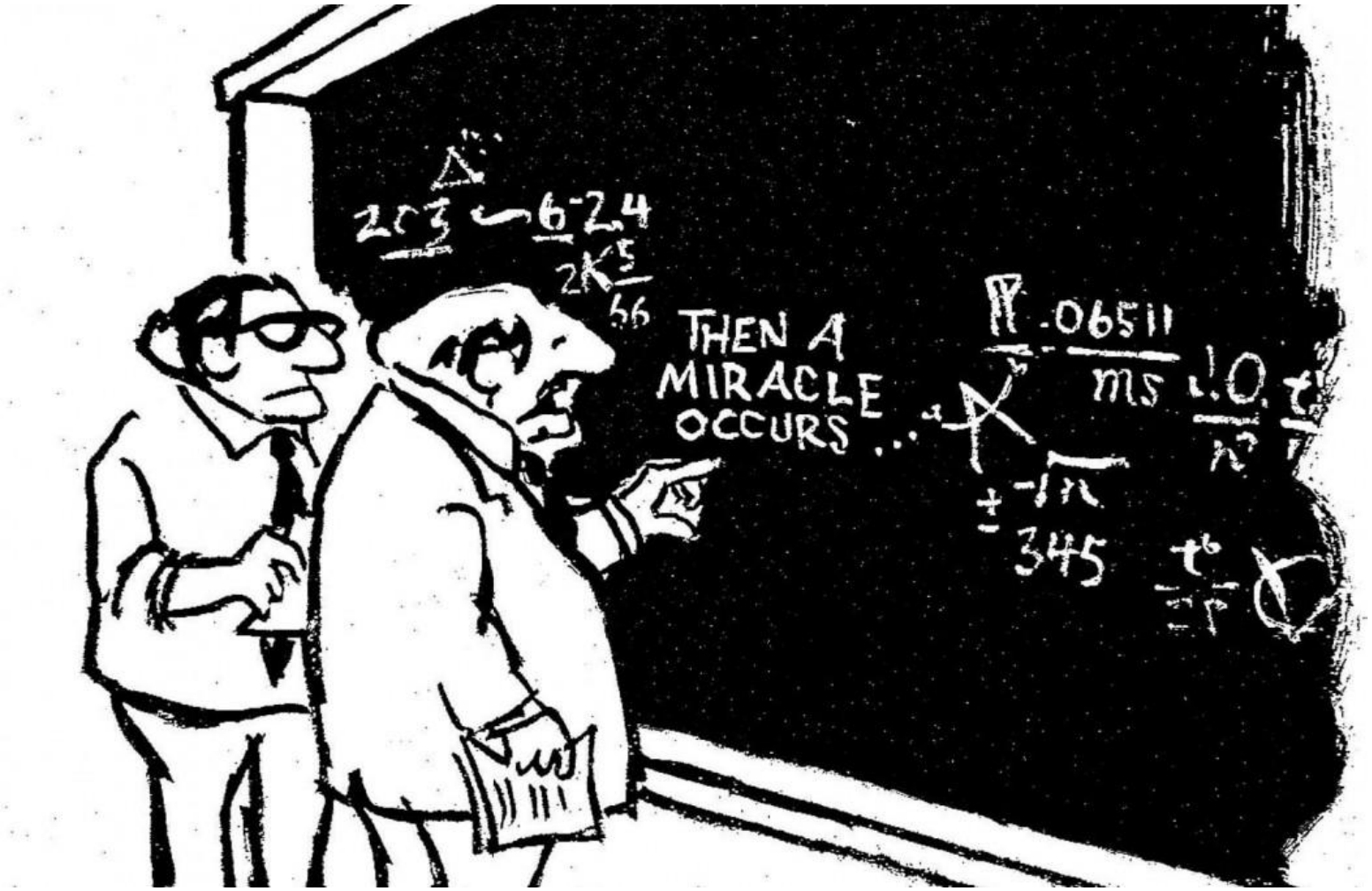
- It is closely related to AI transparency.
- Algorithmic explanation is more related to individual predictions.

Algorithmic interpretability

Algorithmic **interpretability**/**explainability**:

- is especially important for automatic medical diagnostic software.
- relates to the legitimacy of decisions in social/business systems.
- leads to **Accuracy vs. Explainability tradeoff** in various fields of applications.

Algorithmic interpretability



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ML algorithm interpretability

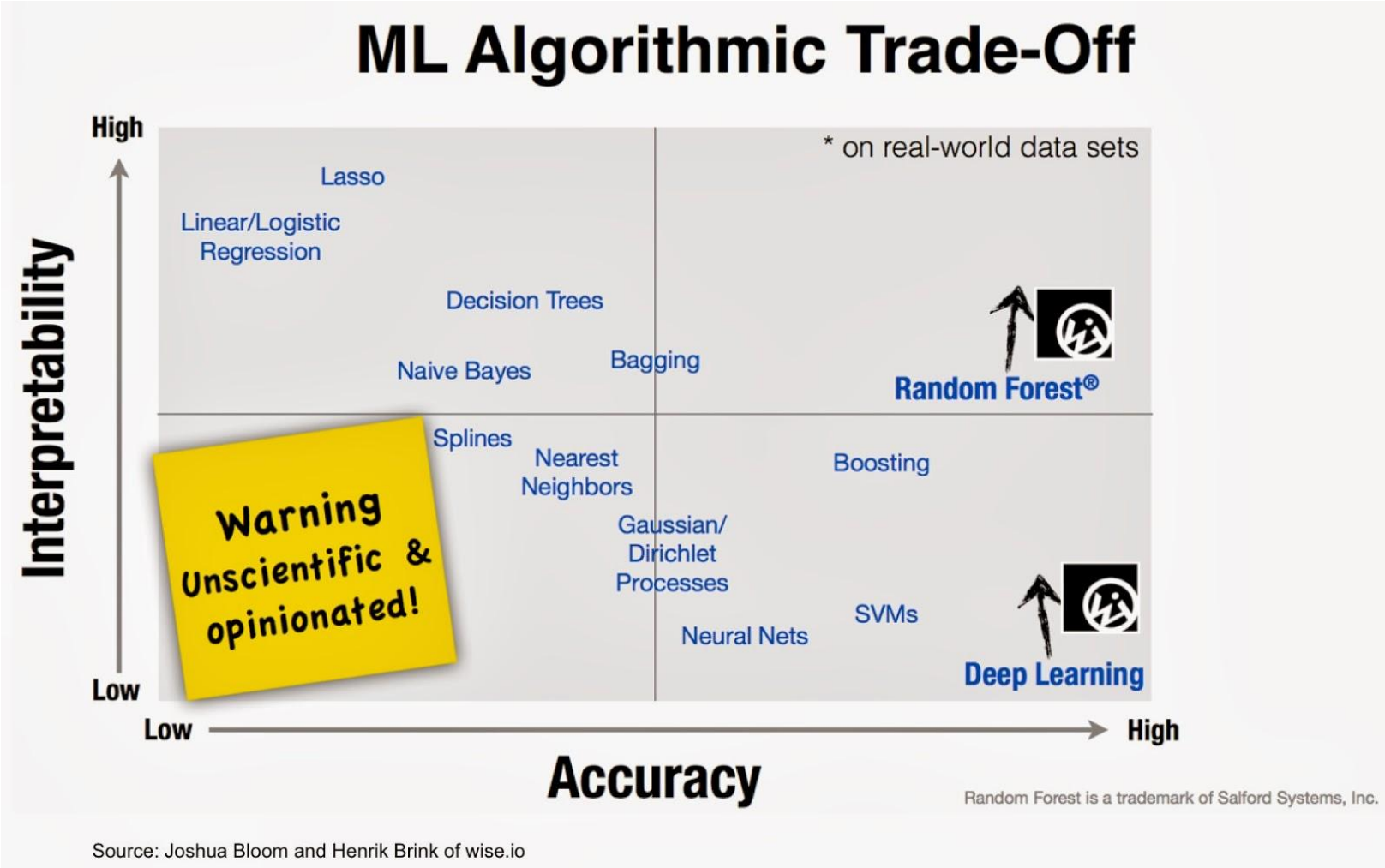
Interpretable ML algorithms:

- Decision forests, linear models, Naïve Bayes, and KNN classifiers

Non-interpretable ML algorithms:

- Deep neural networks, SVM, kernel methods, etc.

ML algorithm interpretability



Source: Joshua Bloom and Henrik Brink of wise.io

Principle of accountable algorithms

1. **Responsibility**

   Who is responsible, if users are harmed?

2. **Explainability**

   How much of the algorithm code and data will be disclosed?

3. **Accuracy**

   Sources of error and their effect? Worst case scenario?

Principle of accountable algorithms

4. Auditability

Public audit? Communication with outside parties?

5. Fairness

Potential damages to different (social) groups by your algorithms?

Ethical cases of ML algorithms

- gender-biased results (**discrimination**)
- racist outcome – classification of black people as „gorillas“ (**discrimination**, **fairness**)
- resume filtering based on age and sex in HR industries (**discrimination, fairness**)
- invisible calculation of credit score (**transparency, accountability**)

Ethical cases of ML algorithms

- data brokers (**confidentiality**)
- Uber taxis price forming (**transparency, fairness**)
- predictive policing (**discrimination**, **fairness**)
- personal and psychological profiling (**privacy, discrimination, confidentiality**)

What are the consequences?

What can happen, if we **do not oppose biases** in ML systems?

- Businesses will use biased datasets for greater profits.
- ML developers will apply evaluation metrics which can amplify biases – gender or race specific.
- The wide application of ML algorithms will strengthen bias and polarization in society.
- Social tension and distrust to AI and technologies will arise.

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